

HK Express: Ensuring high customer satisfaction with Vertex AI

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HK Express leverages Vertex AI notebooks to store, organize, analyze, and visualize data, supporting all business units across the company with accurate, data-led insights.

ABOUT HK EXPRESS



About HK Express

HK Express is an IATA Operational Safety Audit (IOSA) registered airline. Operating as one of the world's most punctual and safest airlines, HK Express is ranked third globally and first in Asia for on-time arrival performance among low-cost carriers by Cirium, a leading aviation analytics firm. It has also received a "7-star safety rating"—the highest honor from [airlineratings.com](#), a renowned independent airline rating review agency worldwide.

As a member of the Cathay Group, HK Express has successfully empowered over 22 million

Google Cloud results

- Manages customer feedback

Experiences
41% efficiency

from
across the
company
with a
lean team

gain in analysis

- Delivers accurate and granular feedback for business units to act upon quickly
- Enables a seamless pipeline with the flexibility of tapping on other open-source generative AI models

passengers to travel on their own terms since transforming into a low-cost carrier. It is widely trusted and favored by its customers.

Industries: Transportation & Logistics

Location: Hong Kong

Storage (<https://cloud.google.com/storage/>),
Google Cloud (<https://cloud.google.com/>)
Cloud Functions

BigQuery (<https://cloud.google.com/bigquery/>)
Data Fusion

Cloud Functions

(<https://cloud.google.com/functions/>)

Cloud Storage (<https://cloud.google.com/storage/>)

Vertex AI (<https://cloud.google.com/vertex-ai/>)

Cloud AutoML (<https://cloud.google.com/automl/>)

Cloud Data Fusion

(<https://cloud.google.com/data-fusion>)

Vertex AI Notebooks

(<https://cloud.google.com/vertex-ai-notebooks>)

As Hong Kong's "one-and-only" low-cost airline, HK Express (<https://www.hkexpress.com/ja-jp/search/>) is focused on more than just delivering flights at unbeatable prices. The company places high priority on passenger safety and is always on the move to deliver a best-in-class service, charting new paths to passenger satisfaction. That's why the airline

worked with Google Cloud (<https://cloud.google.com/>) on a strategy to deploy Vertex AI (<https://cloud.google.com/vertex-ai>) as part of its integrated cloud solution.

In particular, HK Express wanted to empower its lean data science team with a platform that would process customer feedback surveys, quickly turning data into actionable insights. The airline needed a way to ingest, manage, and make sense of the large amount of varied data that comes through from the different business units daily.

Every month, the Journey Experience Design (JED) team receives 10,000 or more surveys for processing. The team has to then manually go through each one, extract and classify every surface comment into the relevant touch points.

"The manual process is not only cumbersome and time consuming, but also allows room for human error," explains Andy Luk, Head of Digital Transformation and Insights at HK Express. To address this, the JED team has tapped the support of the data science team, which has automated the process using Vertex AI PaLM2. "By leveraging Google Cloud to deploy our machine learning solution built on Vertex AI notebooks, we are able to quickly and accurately analyze the data we receive and categorize them into the relevant data tables," he explains.

The multi-language and classification capabilities of Vertex AI PaLM2 helps the JED team extract information and make sense of them, assisting with

customer feedback survey analysis. "Vertex AI PaLM2 categorizes the touchpoints, topics, and subtopics mentioned in each survey, extract relevant quotes that correspond to categorized topics, and can extract additional information like specific meals mentioned or names. It can even do sentiment analysis for each topic in each survey," says Leo Mok, Senior Manager, Data Science & AI at HK Express.

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Building a secure and efficient data pipeline

While customer feedback surveys may seem straightforward, it is not always the case. "For example, in the same survey response, a customer may praise a flight attendant for their good service and say that the in-flight meals were not hot enough," explains Mok. The task for the data team would then be to leverage Vertex AI PaLM2 to extract the two quotes and put them into the relevant buckets. The first, is on cabin crew attitude, and the second, for the in-flight food experience. After this first round of classification, a second round of analysis is then run again to be more specific. "What kind of attitude, is it positive or negative? And we also need to narrow down on the food experience by determining which sentiments go with which specific food on the menu," says Mok.

The team leverages BigQuery (<https://cloud.google.com/bigquery>) to store the topics and subtopics which are provided by the users, and subsequently visualizes the data. Every week, the team utilizes insights from feedback to finetune the topic and subtopic lists on BigQuery and the machine learning (ML) prompt, improving the ML model that runs on Vertex AI over time. "We keep modifying the prompts and topic definitions, and over time, this improves our efficiency gains in terms of time and accuracy," explains Mok, adding that in the two months since deployment, the team has already experienced a 41 percent efficiency improvement.

On the production side, the data science team deploys its solution through the Google Cloud pipeline using Cloud Functions (<https://cloud.google.com/functions>), streamlining its weekly operations. "We used Cloud Functions to automate streaming all data operation tasks through Google Cloud pipelines throughout this project," says Luk. HK Express stores extracted data in Cloud Storage buckets before processing it to be stored in data tables. Then, that data is extracted again, combining the topic definitions with code frames to define the topics for Vertex AI to generate.

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PaLM2

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Mitigating the lack of human resources with AI

When the borders opened for travel to Hong Kong again after the COVID pandemic, the sudden surge in business made it challenging for the airline to keep up with human resources. HK Express faced a common post-pandemic challenge in recruiting enough professionals. "To ease the workload, we leveraged AI to automate many tasks that were previously done manually. One of which is to automate our customer feedback surveys, allowing us to operate at an even more efficient scale, keeping customer satisfaction at the highest level," says Luk.

Meanwhile, the company continues to hire data experts who will be able to focus on more business critical tasks such as fine-tuning the ML models, instead of the manual operations that would otherwise take up most of their time. "With Google Cloud and Vertex AI, a lean data and JED team can now manage the survey feedback that will serve for all different business units across the entire company," says Mok.

Shaping the future of the airline industry with generative AI solutions

Using Vertex AI to improve its customer feedback surveys is only a piece of the puzzle to a much larger picture of what HK Express is doing. With [Gemini](#)

(<https://cloud.google.com/blog/products/ai-machine-learning/bringing-gemini-to-organizations-everywhere>)

set to make its entrance, the team is already making plans to implement its capabilities into its ecosystem.

"The great thing about Vertex AI is that it doesn't constrain us to just Google Cloud in-house models. We have the freedom to choose different models based on the different use-cases," shares Luk. This enables the team to further explore using generative AI technology in other survey formats such as pop-up digital surveys.

As next steps, HK Express plans to explore the Large Language Model technology that Gemini will bring, and use it to analyze the conversations that their agents have with customers on their customer service chat platform. This will enable us to improve the satisfaction rate and also speed up issue resolution.

"The possibilities are endless, and I believe there is much more to be done with Google Cloud and its host of generative AI capabilities. By pairing the right technologies and expertise, we look forward to achieving many more milestones together," says Luk.

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